GENERAL QUESTIONS

1. On the fifth clock pulse, a 4-bit Johnson sequence is $Q_0 = 0$, $Q_1 = 1$, $Q_2 = 1$, and $Q_3 = 1$. On the sixth clock pulse, the sequence is _____.

   A. $Q_0 = 1, Q_1 = 0, Q_2 = 0, Q_3 = 0$

   B. $Q_0 = 1, Q_1 = 1, Q_2 = 1, Q_3 = 0$

   C. $Q_0 = 0, Q_1 = 0, Q_2 = 1, Q_3 = 1$

   D. $Q_0 = 0, Q_1 = 0, Q_2 = 0, Q_3 = 1$

   Answer: Option **C**

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2. The bit sequence 0010 is serially entered (right-most bit first) into a 4-bit parallel out shift register that is initially clear. What are the Q outputs after two clock pulses?

   A. 0000

   B. 0010

   C. 1000

   D. 1111

   Answer: Option **C**

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3. What is a shift register that will accept a parallel input, or a bidirectional serial load and internal shift features, called?

   A. tristate

   B. end around

   C. universal

   D. conversion

   Answer: Option **C**

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4. On the third clock pulse, a 4-bit Johnson sequence is $Q_0 = 1$, $Q_1 = 1$, $Q_2 = 1$, and $Q_3 = 0$. On the fourth clock pulse, the sequence is _____.

   A. $Q_0 = 1, Q_1 = 1, Q_2 = 1, Q_3 = 1$

   B. $Q_0 = 1, Q_1 = 1, Q_2 = 0, Q_3 = 0$

   C. $Q_0 = 1, Q_1 = 0, Q_2 = 0, Q_3 = 0$

   D. $Q_0 = 0, Q_1 = 0, Q_2 = 0, Q_3 = 0$

   Answer: Option **A**

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5. A bidirectional 4-bit shift register is storing the nibble 1101. Its $\overline{\text{RIGHT/LEFT}}$ input is HIGH. The nibble 1011 is waiting to be entered on the serial data-input line. After three clock pulses, the shift register is storing _____.

   A. 1101

   B. 0111

C. 0001

D. 1110

Answer: Option **B**

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6. How can parallel data be taken out of a shift register simultaneously?

A. Use the *Q* output of the first FF.

B. Use the *Q* output of the last FF.

C. Tie all of the *Q* outputs together.

D. Use the *Q* output of each FF.

Answer: Option **D**

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7. What is meant by parallel load of a shift register?

A. All FFs are preset with data.

B. Each FF is loaded with data, one at a time.

Answer: Option **A**

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8. What does the output enable do on the 74395A chip?

A. It determines when data can be loaded.

B. It forces all outputs to go HIGH.

C. It forces all outputs to go LOW.

D. It activates the three-state buffer.

Answer: Option **D**

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9. To operate correctly, starting a ring shift counter requires:

A. clearing all the flip-flops

B. presetting one flip-flop and clearing all others

C. clearing one flip-flop and presetting all others

D. presetting all the flip-flops

Answer: Option **B**

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10. In a 6-bit Johnson counter sequence there are a total of how many states, or bit patterns?

A. 2

B. 6

C. 12

D. 24

Answer: Option **C**

11. A modulus-12 ring counter requires a minimum of _____.
    - **A.** 10 flip-flops
    - **B.** 12 flip-flops
    - **C.** 6 flip-flops
    - **D.** 2 flip-flops

    Answer: Option **B**

12. Stepper motors have become popular in digital automation systems because _____.
    - **A.** of their low cost
    - **B.** they are driven by sequential digital signals
    - **C.** they can be used to provide repetitive mechanical movement
    - **D.** they are driven by sequential digital signals and can be used to provide repetitive mechanical movement

    Answer: Option **D**

13. The group of bits 11001 is serially shifted (right-most bit first) into a 5-bit parallel output shift register with an initial state 01110. After three clock pulses, the register contains _____.
    - **A.** 01110
    - **B.** 00001
    - **C.** 00101
    - **D.** 00110

    Answer: Option **C**

14. Assume that a 4-bit serial in/serial out shift register is initially clear. We wish to store the nibble 1100. What will be the 4-bit pattern after the second clock pulse? (Right-most bit first.)
    - **A.** 1100
    - **B.** 0011
    - **C.** 0000
    - **D.** 1111

    Answer: Option **C**

15. A serial in/parallel out, 4-bit shift register initially contains all 1s. The data nibble 0111 is waiting to enter. After four clock pulses, the register contains _____.
    - **A.** 0000
    - **B.** 1111

C. 0111

D. 1000

Answer: Option **C**

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16. With a 200 kHz clock frequency, eight bits can be serially entered into a shift register in _____.

A. 4 µs

B. 40 µs

C. 400 µs

D. 40 ms

Answer: Option **B**

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17. An 8-bit serial in/serial out shift register is used with a clock frequency of 2 MHz to achieve a time delay (t_d) of _____.

A. 16 µs

B. 8 µs

C. 4 µs

D. 2 µs

Answer: Option **C**

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18. A sequence of equally spaced timing pulses may be easily generated by which type of counter circuit?

A. ring shift

B. clock

C. Johnson

D. binary

Answer: Option **A**

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19. The bit sequence 10011100 is serially entered (right-most bit first) into an 8-bit parallel out shift register that is initially clear. What are the Q outputs after four clock pulses?

A. 10011100

B. 11000000

C. 00001100

D. 11110000

Answer: Option **B**

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20. If an 8-bit ring counter has an initial state 10111110, what is the state after the fourth clock pulse?

A. 11101011

B. 00010111

C. 11110000

D. 00000000

Answer: Option **A**

---

21. How would a latch circuit be used in a microprocessor system?

A. as transportation for Intel employees

B. for a group of data that is the same

C. as a set of common connections for transfer of data

Answer: Option **C**

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22. A 4-bit shift register that receives 4 bits of parallel data will shift to the _____ by _____ position(s) for each clock pulse.

A. right, one

B. right, two

C. left, one

D. left, three

Answer: Option **A**

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23. How many clock pulses will be required to completely load serially a 5-bit shift register?

A. 2

B. 3

C. 4

D. 5

Answer: Option **D**

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24. How is a strobe signal used when serially loading a shift register?

A. to turn the register on and off

B. to control the number of clocks

C. to determine which output *Q*s are used

D. to determine the FFs that will be used

Answer: Option **B**

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25. An 8-bit serial in/serial out shift register is used with a clock frequency of 150 kHz. What is the time delay between the serial input and the Q_3 output?

A. 1.67 μs

B. 26.67 μs

C. 26.7 ms

D. 267 ms

Answer: Option **B**

---

26. What are the three output conditions of a three-state buffer?

A. HIGH, LOW, float

B. 1, 0, float

C. both of the above

D. neither of the above

Answer: Option **C**

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27. The primary purpose of a three-state buffer is usually:

A. to provide isolation between the input device and the data bus

B. to provide the sink or source current required by any device connected to its output without loading down the output device

C. temporary data storage

D. to control data flow

Answer: Option **A**

---

28. What is the difference between a ring shift counter and a Johnson shift counter?

A. There is no difference.

B. A ring is faster.

C. The feedback is reversed.

D. The Johnson is faster.

Answer: Option **C**

---

29. What is a recirculating register?

A. serial out connected to serial in

B. all *Q* outputs connected together

C. a register that can be used over again

Answer: Option **A**

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30. When is it important to use a three-state buffer?

A. when two or more outputs are connected to the same input

B. when all outputs are normally HIGH

C. when all outputs are normally LOW

D. when two or more outputs are connected to two or more inputs

Answer: Option **A**

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31. A bidirectional 4-bit shift register is storing the nibble 1110. Its $\overline{\text{RIGHT}}/\text{LEFT}$ input is LOW. The nibble 0111 is waiting to be entered on the serial data-input line. After two clock pulses, the shift register is storing _____.

A. 1110

B. 0111

C. 1000

D. 1001

Answer: Option **D**

---

32. In a parallel in/parallel out shift register, $D_0 = 1$, $D_1 = 1$, $D_2 = 1$, and $D_3 = 0$. After three clock pulses, the data outputs are _____.

A. 1110

B. 0001

C. 1100

D. 1000

Answer: Option **B**

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33. The group of bits 10110111 is serially shifted (right-most bit first) into an 8-bit parallel output shift register with an initial state 11110000. After two clock pulses, the register contains _____.

A. 10111000

B. 10110111

C. 11110000

D. 11111100

Answer: Option **D**

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34. By adding recirculating lines to a 4-bit parallel-in, serial-out shift register, it becomes a _____, _____, and _____-out register.

A. parallel-in, serial, parallel

B. serial-in, parallel, serial

C. series-parallel-in, series, parallel

D. bidirectional in, parallel, series

Answer: Option **A**

35. What type of register would have a complete binary number shifted in one bit at a time and have all the stored bits shifted out one at a time?

A. parallel-in, parallel-out

B. parallel-in, serial-out

C. serial-in, parallel-out

D. serial-in, serial-out

Answer: Option **C**

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36. When an 8-bit serial in/serial out shift register is used for a 20 μs time delay, the clock frequency is _____.

A. 40 kHz

B. 50 kHz

C. 400 kHz

D. 500 kHz

Answer: Option **C**

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37. Ring shift and Johnson counters are:

A. synchronous counters

B. aynchronous counters

C. true binary counters

D. synchronous and true binary counters

Answer: Option **A**

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38. What is the difference between a shift-right register and a shift-left register?

A. There is no difference.

B. the direction of the shift

Answer: Option **B**

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39. What is a transceiver circuit?

A. a buffer that transfers data from input to output

B. a buffer that transfers data from output to input

C. a buffer that can operate in both directions

Answer: Option **C**

---

40. A 74HC195 4-bit parallel access shift register can be used for _____.

A. serial in/serial out operation

B. serial in/parallel out operation

C. parallel in/serial out operation

D. all of the above

Answer: Option **D**

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41. Which type of device may be used to interface a parallel data format with external equipment's serial format?

A. key matrix

B. UART

C. memory chip

D. series in, parallel out

Answer: Option **B**

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42. What is the function of a buffer circuit?

A. to provide an output that is inverted from that on the input

B. to provide an output that is equal to its input

C. to clean up the input

D. to clean up the output

Answer: Option **B**

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43. What is the preset condition for a ring shift counter?

A. all FFs set to 1

B. all FFs cleared to 0

C. a single 0, the rest 1

D. a single 1, the rest 0

Answer: Option **D**

---

44. Which is not characteristic of a shift register?

A. Serial in/parallel in

B. Serial in/parallel out

C. Parallel in/serial out

D. Parallel in/parallel out

Answer: Option **A**

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45. To keep output data accurate, 4-bit series-in, parallel-out shift registers employ a _____.

A. divide-by-4 clock pulse

B. sequence generator

C. strobe line

D. multiplexer

Answer: Option **C**

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46. With a 50 kHz clock frequency, six bits can be serially entered into a shift register in _____.

A. 12 μs

B. 120 μs

C. 12 ms

D. 120 ms

Answer: Option **B**

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47. Another way to connect devices to a shared data bus is to use a _____.

A. circulating gate

B. transceiver

C. bidirectional encoder

D. strobed latch

Answer: Option **B**

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48. To serially shift a nibble (four bits) of data into a shift register, there must be _____.

A. one clock pulse

B. four clock pulses

C. eight clock pulses

D. one clock pulse for each 1 in the data

Answer: Option **B**

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49. Computers operate on data internally in a _____ format.

A. tristate

B. universal

C. parallel

D. serial

Answer: Option **C**

---

50. In a 4-bit Johnson counter sequence there are a total of how many states, or bit patterns?

A. 1

B. 2

C. 4

D. 8

Answer: Option **D**

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51. If a 10-bit ring counter has an initial state 1101000000, what is the state after the second clock pulse?

A. 1101000000

B. 0011010000

C. 1100000000

D. 0000000000

Answer: Option **B**

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52. How much storage capacity does each stage in a shift register represent?

A. One bit

B. Two bits

C. Four bits (one nibble)

D. Eight bits (one byte)

Answer: Option **A**

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53. When the output of a tristate shift register is disabled, the output level is placed in a:

A. float state

B. LOW state

C. high-impedance state

D. float or high-impedance state

Answer: Option **D**
